Project Design Phase
Solution Architecture

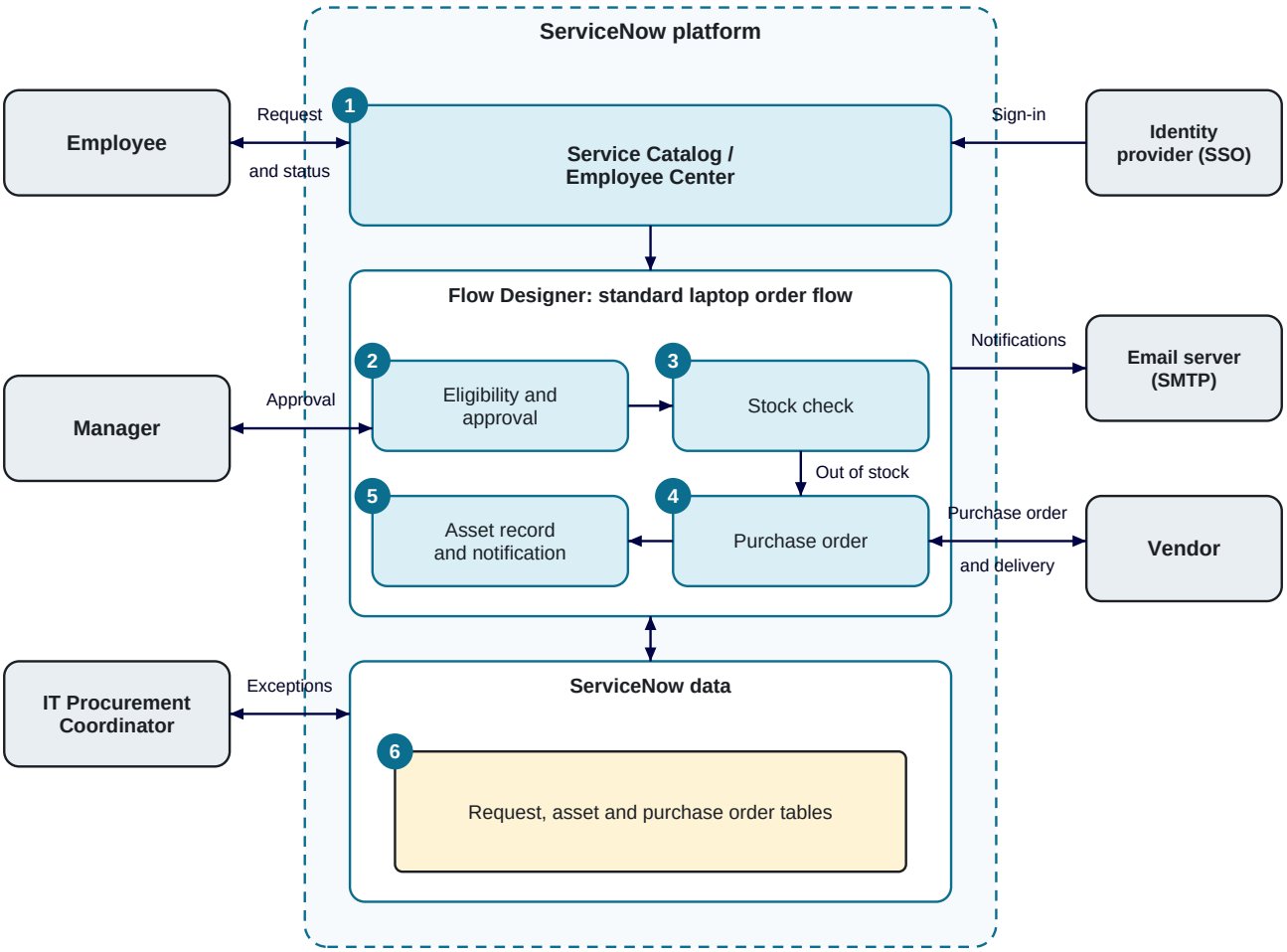
Date	31 January 2025
Team ID	
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process, with many sub-processes, that bridges the gap between business problems and technology solutions. For this project, it shows how the standard laptop order is handled in ServiceNow, from the employee request to delivery.

Goals:

- Find the best technology solution to solve the existing business problem of slow, manual laptop ordering.
- Describe the structure, characteristics and behavior of the solution to project stakeholders.
- Define the features, development phases and solution requirements.
- Provide specifications according to which the solution is defined, managed and delivered.



Architecture Components and Flow:

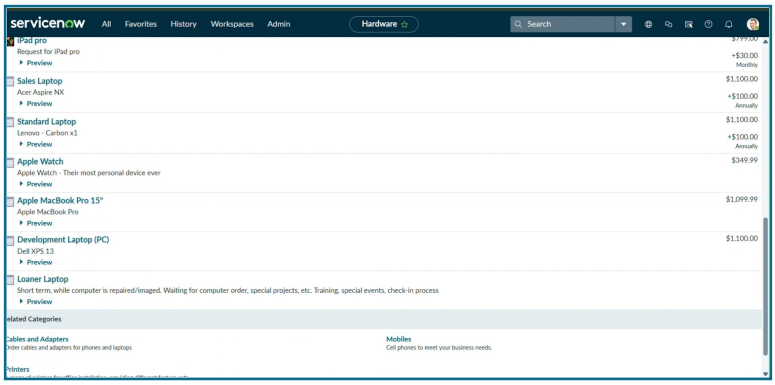
The numbers on the diagram match the components below.

No.	Component	Role in the solution
1	**Service Catalog / Employee Center**	The employee submits one standard laptop request. Details are pre-filled from the user record and the status is visible here.
2	**Eligibility and approval**	Flow Designer checks eligibility, approves automatically for pre-approved roles, or sends a one-click approval task to the manager and escalates it if pending.
3	**Stock check**	The flow checks asset inventory and reserves a laptop when one is available.
4	**Purchase order**	When stock is low, the flow creates a purchase order and sends it to the vendor assigned by region or price.
5	**Asset record and notification**	On delivery the flow creates the asset record, assigns it to the employee and sends notifications at each stage.
6	**ServiceNow data**	Request, asset and purchase order tables store the records used by every step and by reports and dashboards.
-	**External systems**	The identity provider handles sign-in, the email server sends notifications, and the vendor receives purchase orders.

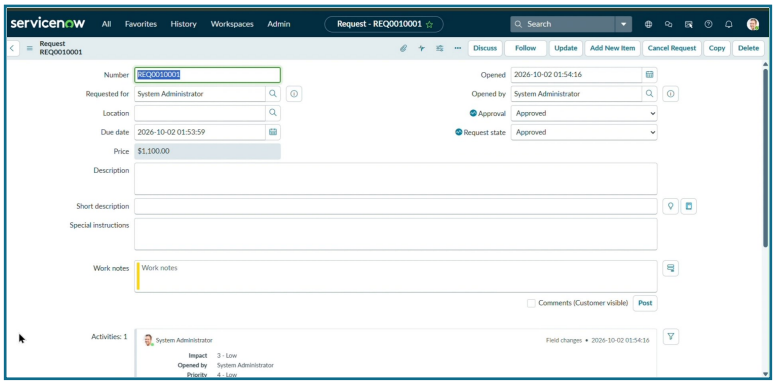
> Flow Designer orchestrates every step inside the ServiceNow platform, so the employee, manager and coordinator work in one system instead of passing the request between email and spreadsheets.

Screenshots

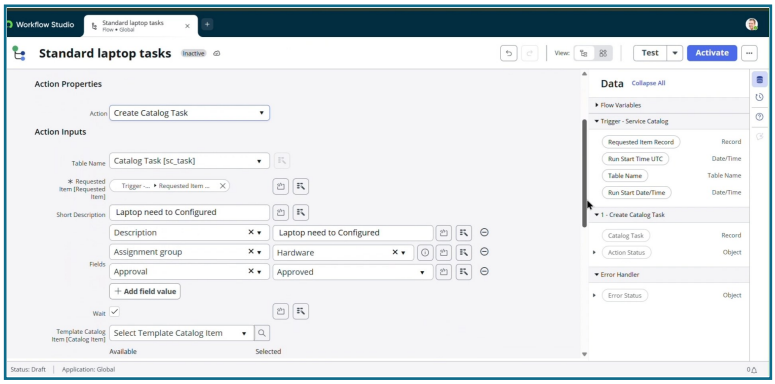
Screenshots from the ServiceNow instance that match components 1 and 2 of the architecture, plus the Flow Designer step that creates the fulfilment task.



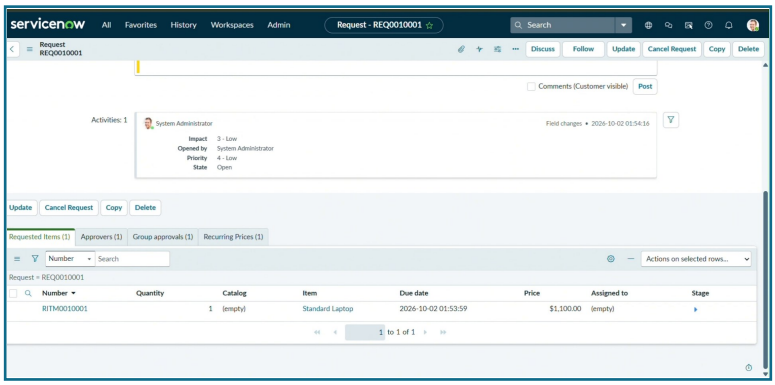
Component 1: Service Catalog with the Standard Laptop item



Component 2: Approval recorded on the request



Flow Designer: Create Catalog Task action in the order flow



Flow output: catalog task SCTASK0010001 created and assigned to Hardware